ML, DL and GenAI

Manaranjan Pradhan

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About Me

- Consulting and training on Big data, AI & Machine Learning.
- An alumni of *IIM, Bangalore*.
- Has about 20+ years of industry experience.
- Has trained 1000+ professionals on Big Data and AI & ML.
- An adjunct faculty at IIM, Bangalore, ISB, Hyderabad and Jio Institute

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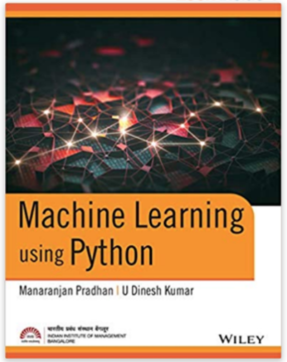
Manaranjan has co-authored the best-selling book Machine Learning using Python

He has published the following machine learning cases in **(HBR) Harvard Business Publishing**:

1. Customer Analytics at Big Basket – Product Recommendations
2. Improving Lead Generation at Eureka Forbes Using Machine Learning Algorithms

Look inside



Machine Learning using Python Paperback – 2019

by U Dinesh Kumar Manaranjan Pradhan (Author)

7 customer reviews

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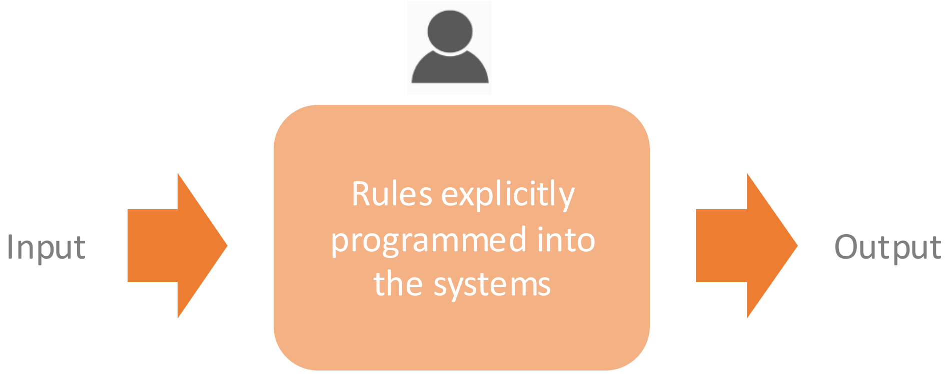
This book is written to provide a strong foundation in machine learning using Python libraries by providing real-life case studies and examples. It covers topics such as foundations of machine learning, introduction to Python, descriptive analytics and predictive analytics. Advanced machine learning concepts such as decision tree learning, random forest, boosting, recommended systems, and text analytics are covered. The book takes a balanced approach between theoretical understanding and practical applications. All the topics include real-world examples and provide step-by-step approach on how to explore, build, evaluate, and optimize machine learning models.

https://www.amazon.in/Machine-Learning-Python-Manaranjan-Pradhan-ebook/dp/B07RLQPNRX

Rule Based or Expert Systems



Rule Based Systems

Static Rules:

- If a user's credit card country points to the US but their IP points to Russia, then the transaction should be blocked.

Velocity Rules:

These rules attempt to **understand user behaviour by looking at set actions over a time period**.

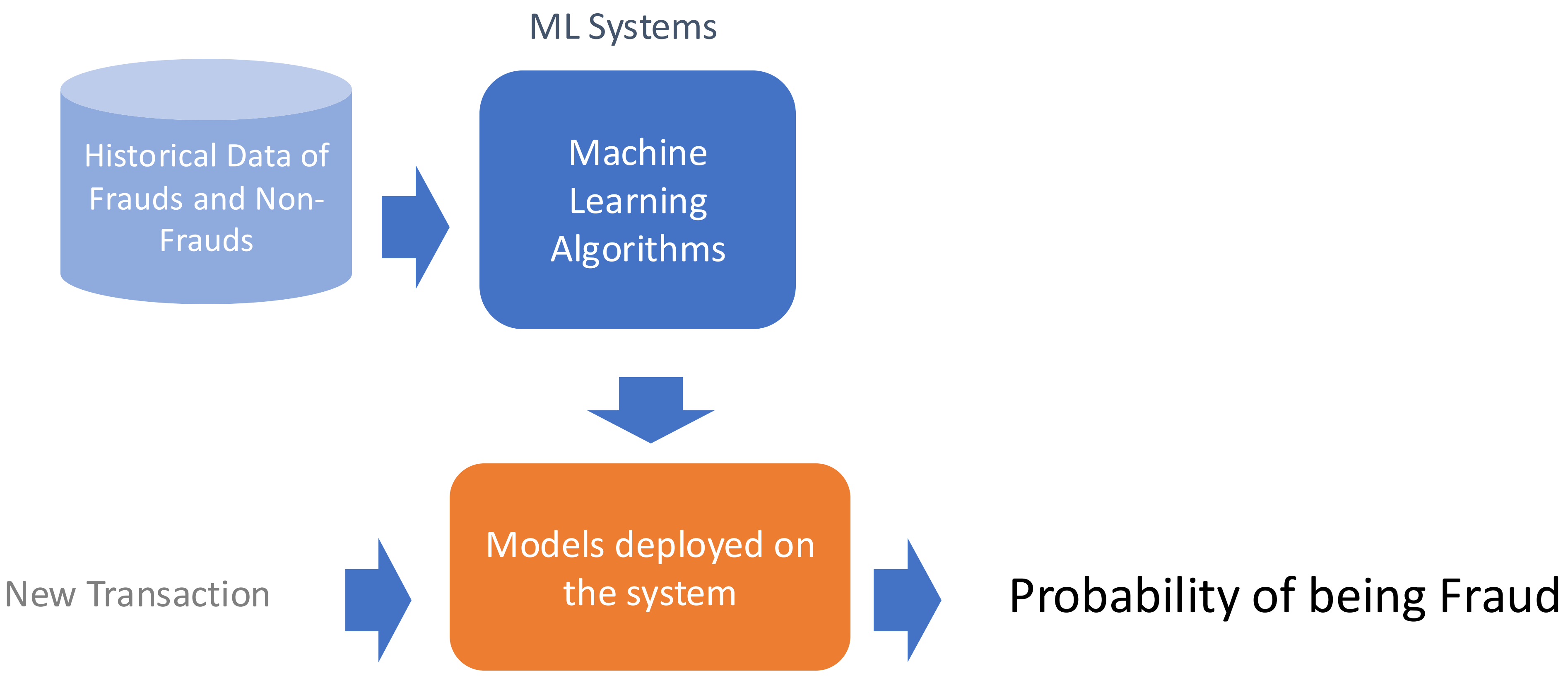
- An increase in spending (more than 200%) over a 24-hour period
- A single user attempting to pay with five different frozen credit cards *within ten minutes* is highly suspicious, as even someone in dire straits would likely stop once they realize one or two of their cards have been frozen or cancelled.

https://seon.io/resources/guides/guide-to-fraud-detection-rules/

Limitation of Rule Based Systems

- Manual Input
- Self Learning / Adapt to changes
- Time Consuming
- Complex Patterns Identification
- Difficult to maintain

Machine Learning Systems

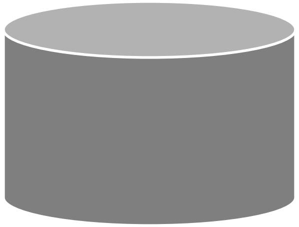


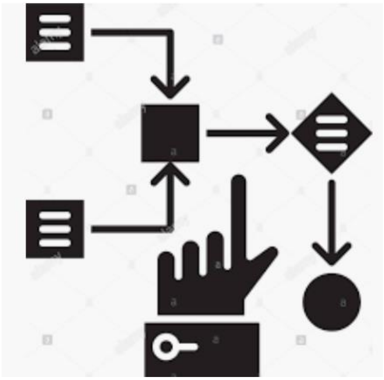
What is Machine Learning?

Machine learning is a field of study that gives computers the ability to learn without explicitly being programmed.

Source: MIT Sloan

Key elements of Machine Learning





$$y = f(x1, x2, x3....)$$

Data

- Past experiences
- Samples representing problem context

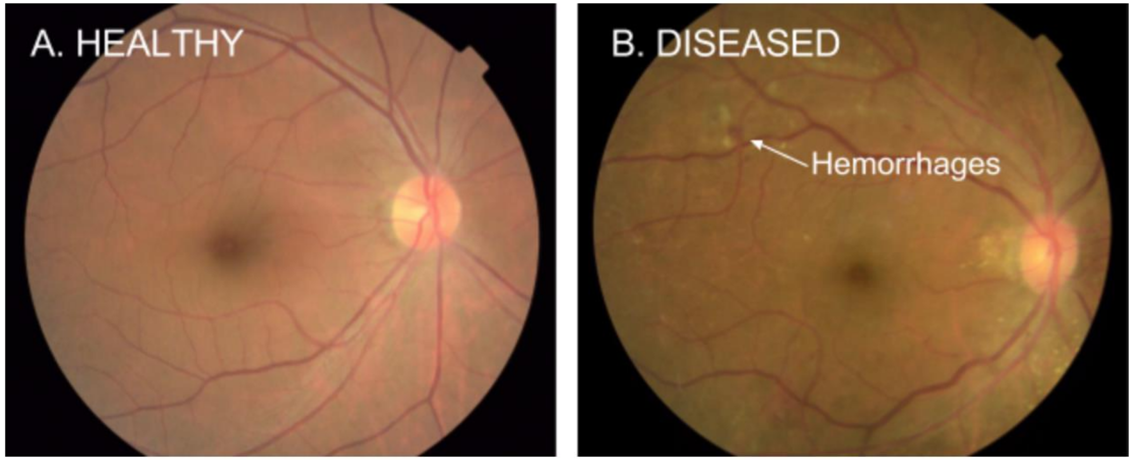
Algorithms

- Machine Learning
- Iteratively goes through the data to find a pattern in the data

Model

- A mathematical expression or set of rules representing the the pattern found in the data

Detection of Diabetic Eye Disease



haemorrhage
/ˈhɛmərɪdʒ/

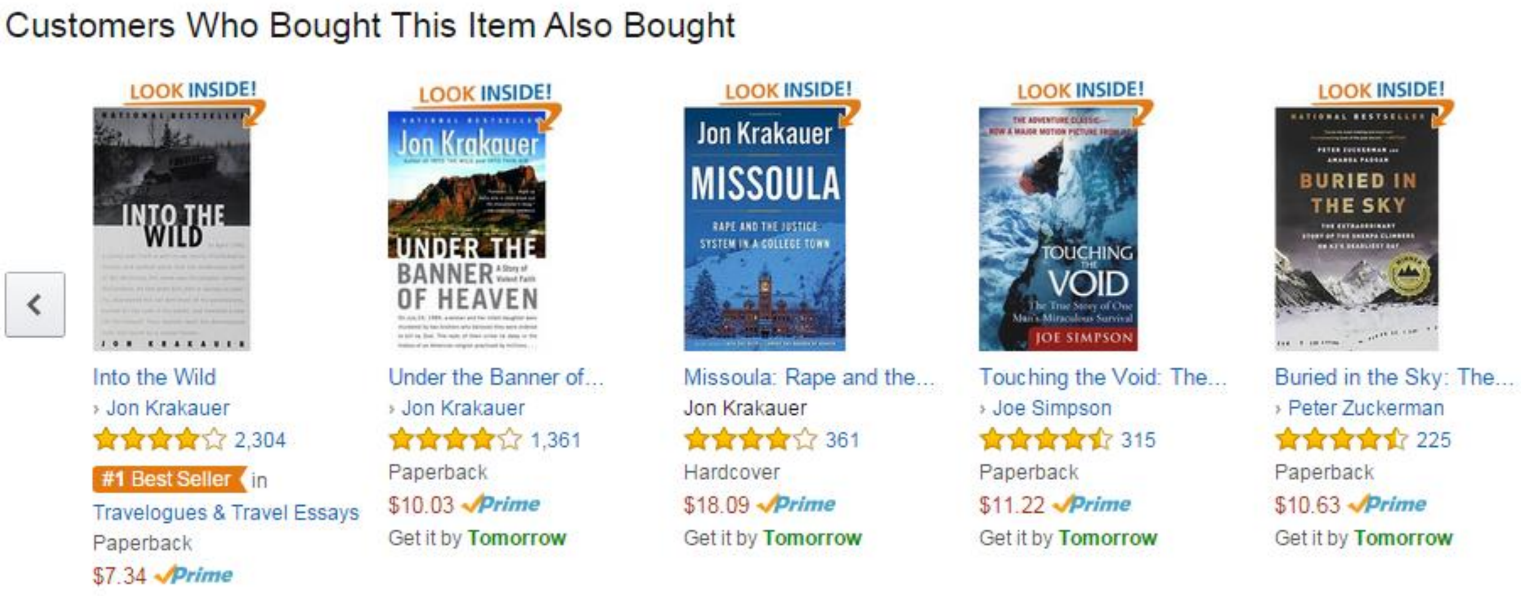
noun
plural noun: **hemorrhages**

1. an escape of blood from a ruptured blood vessel.
"a massive haemorrhage of the brain"

https://research.googleblog.com/2016/11/deep-learning-for-detection-of-diabetic.html

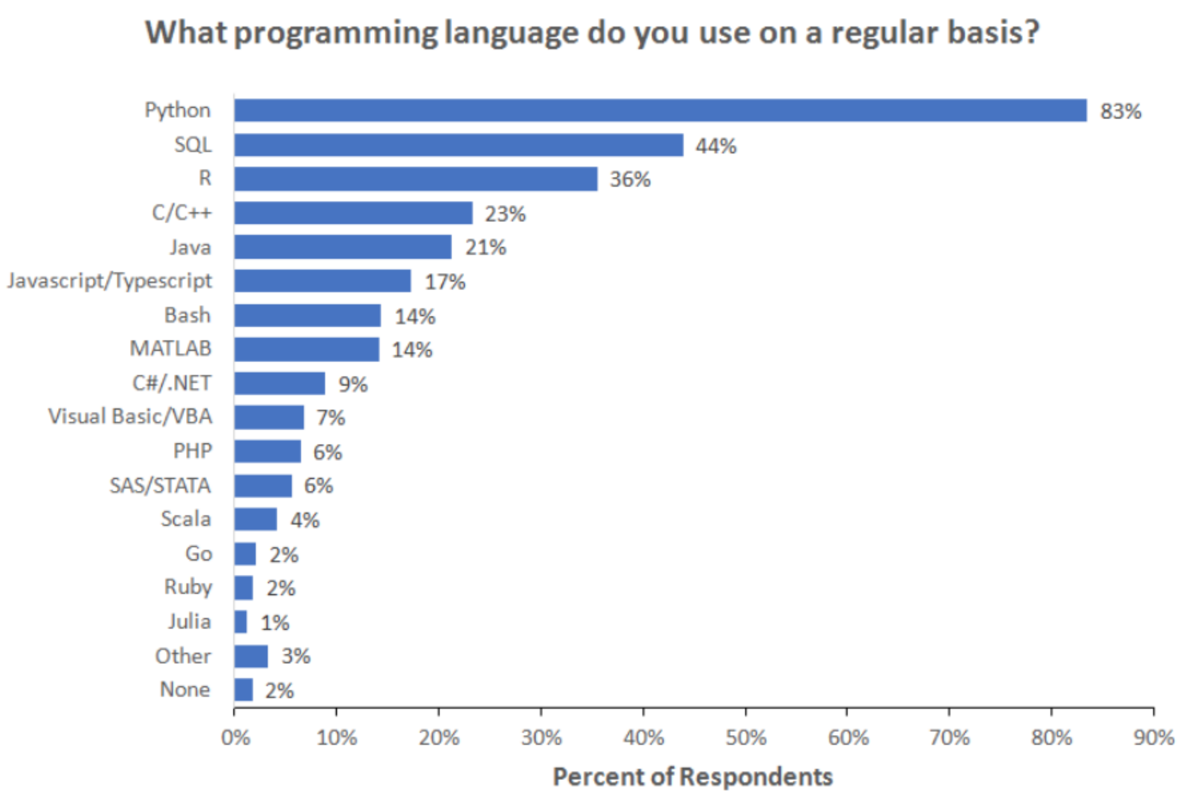
- Google working closely with doctors both in India and the US, created a development dataset of 128,000 images which were each evaluated by 3-7 ophthalmologists from a panel of 54 ophthalmologists.
- Trained a deep neural network to detect referable diabetic retinopathy.
- Then tested the algorithm’s performance on two separate clinical validation sets totaling ~12,000 images, with the majority decision of a panel 7 or 8 U.S. board-certified ophthalmologists serving as the reference standard.
- the algorithm has a F-score (combined sensitivity and specificity metric, with max=1) of 0.95, which is slightly better than the median F-score of the 8 ophthalmologists we consulted (measured at 0.91).

Customers who bought this item also bought…



Why Python?

Language for Machine Learning



Note: Data are from the 2018 Kaggle Machine Learning and Data Science Survey. You can learn more about the study here: http://www.kaggle.com/kaggle/kaggle-survey-2018. A total of 18827 respondents answered the question.

https://businessoverbroadway.com/2019/01/13/programming-languages-most-used-and-recommended-by-data-scientists/

Python Stack For Data Science

Efficient storage of arrays and matrices. Backbone of all scientific calculations and algorithms.

NumPy

Library for scientific computing. Linear algebra, statistical computations, optimization algorithm.

SciPy

Plotting and visualization

seaborn

matplotlib

IP[y]: IPython Interactive Computing

pandas

learn

High-performance, easy-to-use data structures for data manipulation and analysis. Pandas provide the features of dataframe, which is very popular in the area of analytics for data munging, cleaning & transformation.

IDE or Development environment for data analysis in python.

Machine learning library. Collection of ML algorithms.

Development Environment and Tools

Github

The Jupyter Notebook is **a web-based interactive computing platform**.

https://www.jupyter.org/

The Jupyter Notebook is **a web-based interactive computing platform**.

https://www.github.com/

Platforms

ANACONDA

Most popular open-source Python distribution platform

https://www.anaconda.com/products/distribution

colab

Goole Colaboratory is a hosted Jupyter notebook environment that is free to use and requires no setup.

https://colab.research.google.com/

ML Lifecycle

1. Use Case Identification and Problem Formulation
2. Data Collection and Preparation
3. Exploratory Analysis & Feature Engineering
4. Building Machine Learning Models
5. Model Validation and Evaluation
6. Model Deployment

Iterative Steps in Model Development

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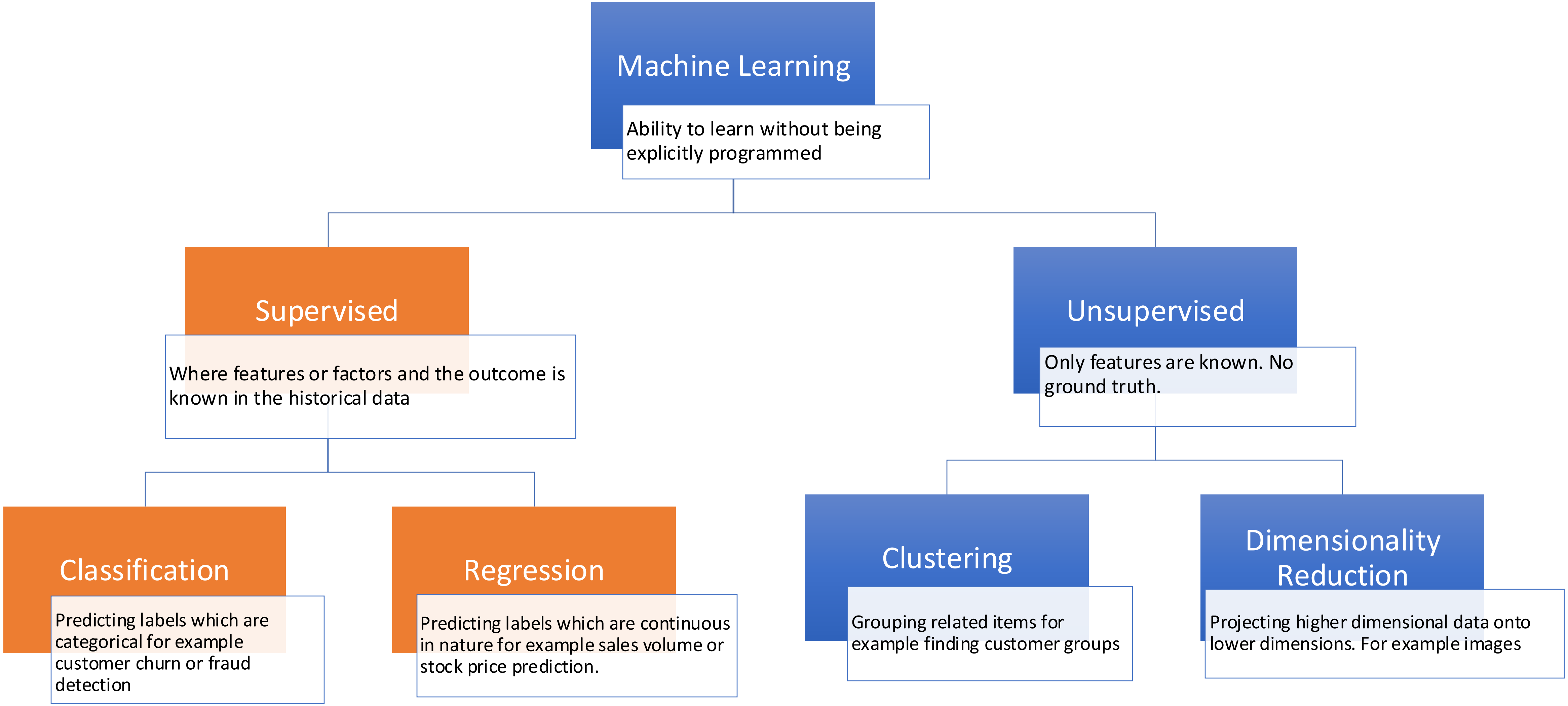
What are different problems that you can solve with ML?

Machine Learning Algorithms



ML Lifecycle

- Use Case Identification and Problem Formulation
- Data Collection and Preparation
- Exploratory Analysis & Feature Engineering
- Building Machine Learning Models
- Model Validation and Evaluation
- Model Deployment
- Model Monitoring

First Model Development Iterative Steps

Continuous Model Update Iterative Steps

Examples of Machine Learning Problems

- What will be the price of a stock in next 3 months given its past performances and the current market outlook?
- What is the estimated demand for a specific product (volume of sale in terms of number of units) in the next quarter given the market conditions and competition?
- What is the likelihood of a customer churning in next 3 months given his/her purchase patterns in the last few months?

Examples of Machine Learning Problems

- What is the likelihood of an employee leaving an organization in next 6 months given his/her performance, behavior and skill demand in the market?
- How to cluster customers together based on their demographics and behavior so that appropriate products or promotions can be targeted to them?
- Which products are customers buying together, which can become candidate for cross selling and up selling?

Examples of Machine Learning Problems

- Identify similar products (movies, books etc.) to recommend to customers based on their preferences shown in the past.
- How to identify anomalies in the systems (machines or hardware) so that preventive maintenance can be scheduled rather than periodic maintenance to avoid catastrophic failures?

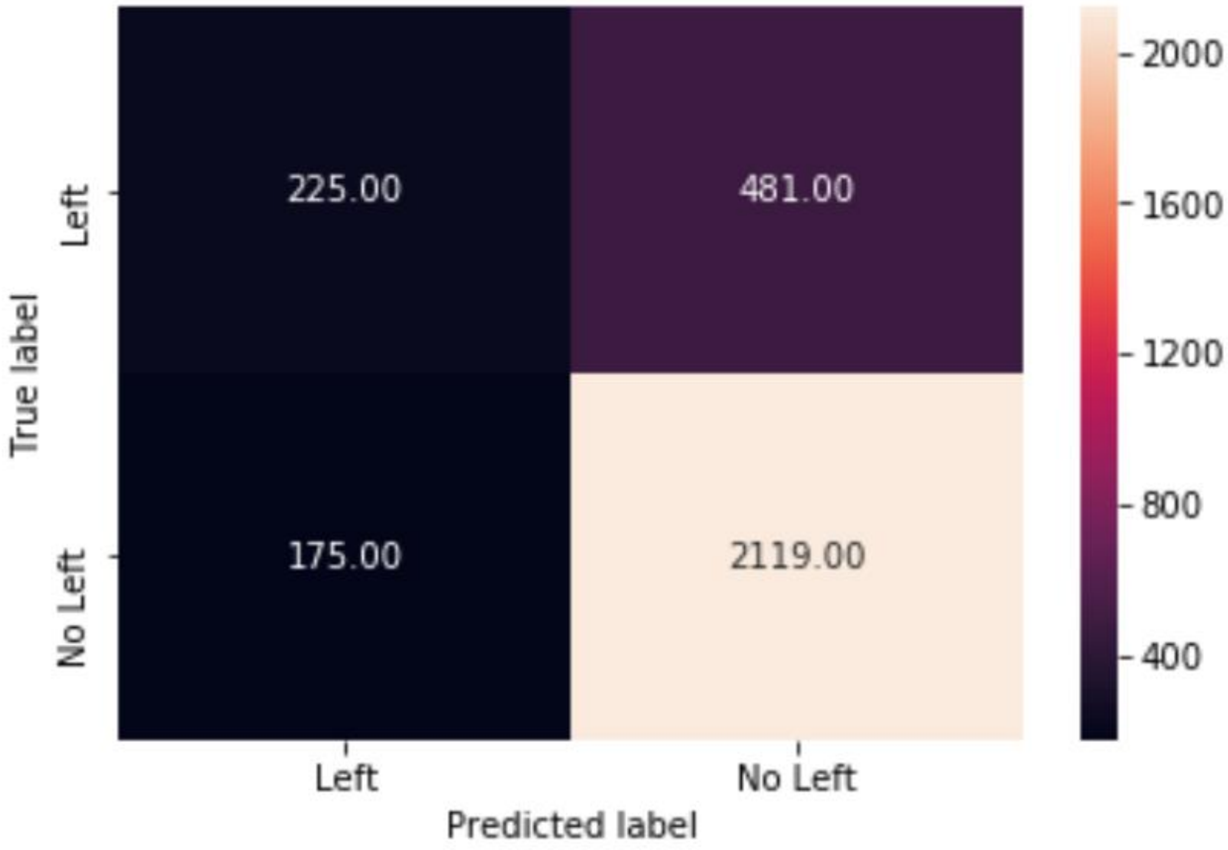
How to estimate the sale price of an used car?

HR Attrition Case Study

We spend huge amount of time and money to hire and nurture their employees. It is a huge loss for companies if employees leave, especially the key resources.

Can we identify the attrition risks and and make policy or employee level intervention to retain those employees or do preventive hiring to minimize the impact.

Confusion Matrix



Classification Metrics

Precision is defined as how many are actual positives out of total number of positives identified by the model and is defined as

$$TPR = (\frac{TP}{TP+FP})$$

True Positive Rate (TPR) or Recall or Sensitivity is how many actual positive are properly identified by the model out of total number actual positive in the test set and is defined as

$$TPR = (\frac{TP}{TP+FN})$$

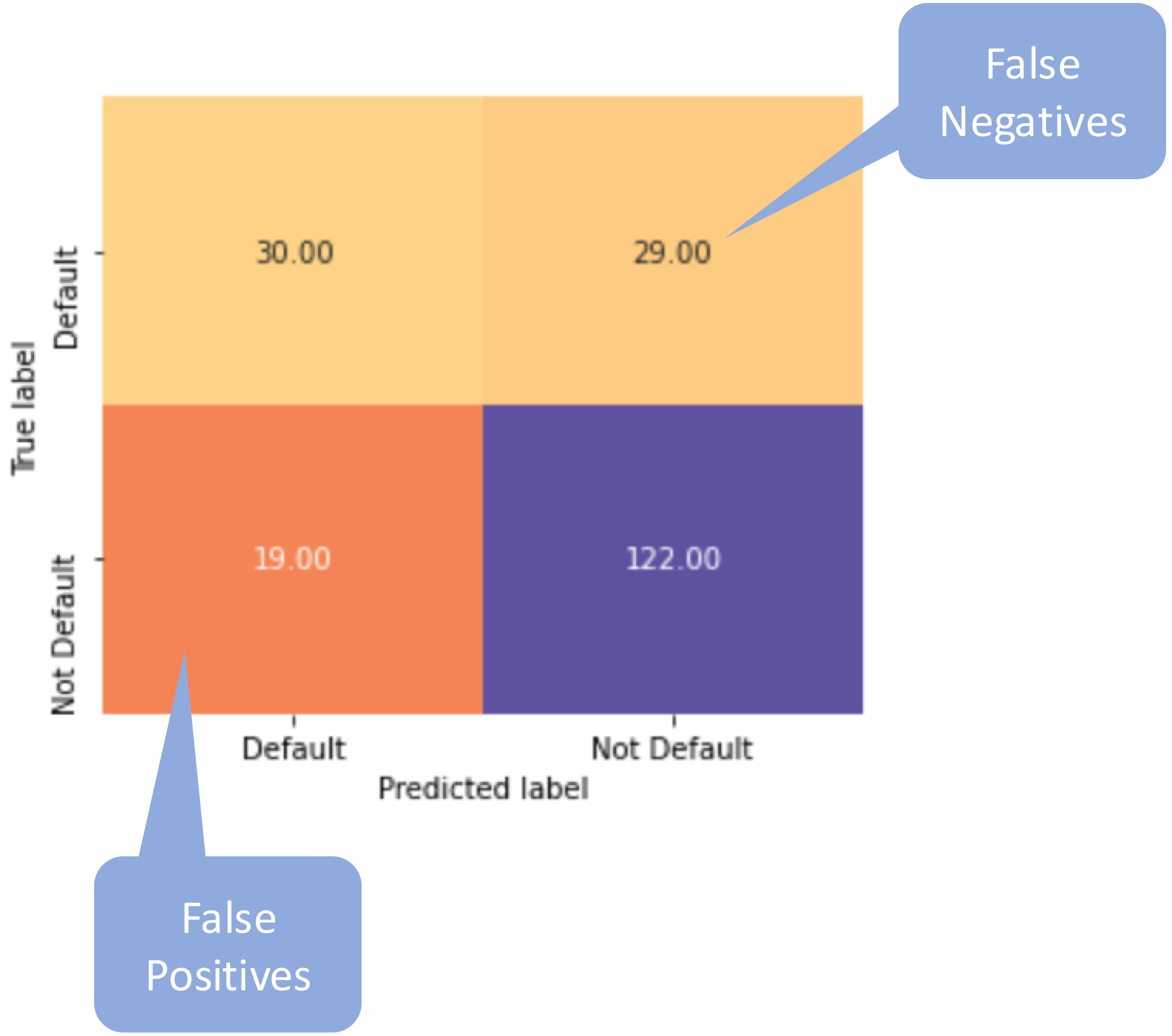
True Negative Rate (TNR) or Specificity is how many are correctly indentified as correct negatives out of all acutal negative present in the test set and is defined as

$$TNR = (\frac{TN}{FP+TN})$$

F-Score (F-Measure) is another measure used in binary logistic regression that combines both precision and recall (harmonic mean of precision and recall) and is given by

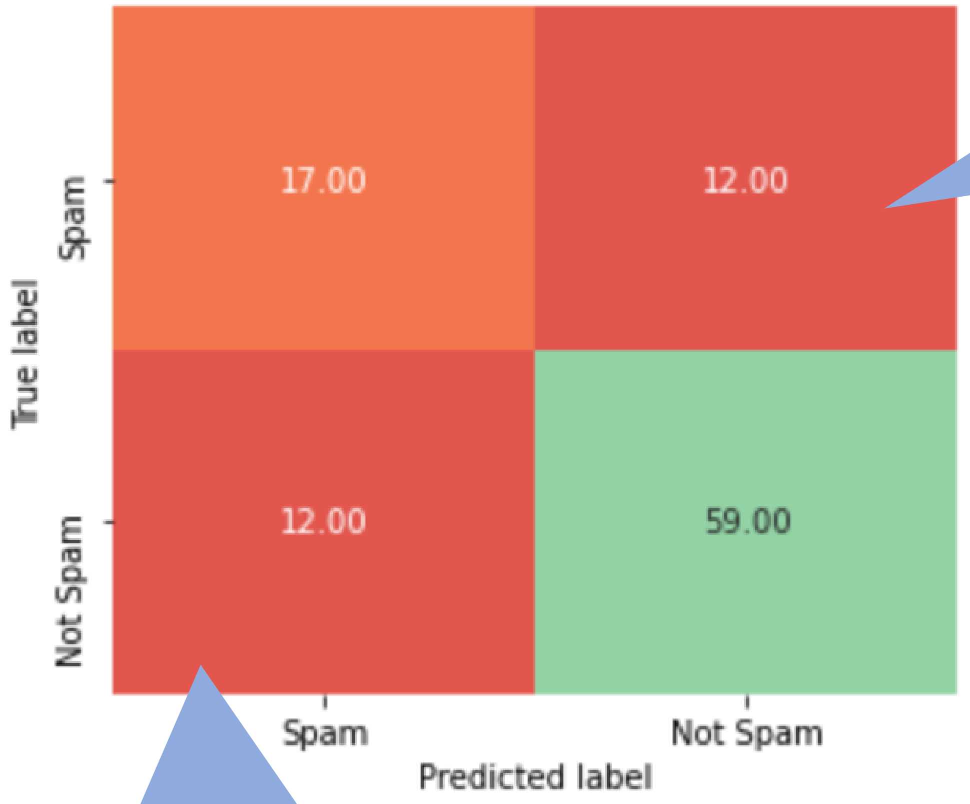
$$F1 - score = (\frac{2xPrecisionxRecall}{Precision+Recall})$$

Cost of Failure: Loan Default



$$\text{Recall} = \frac{TP}{(TP+FN)}$$

Cost of Failure: Spam Detection Model

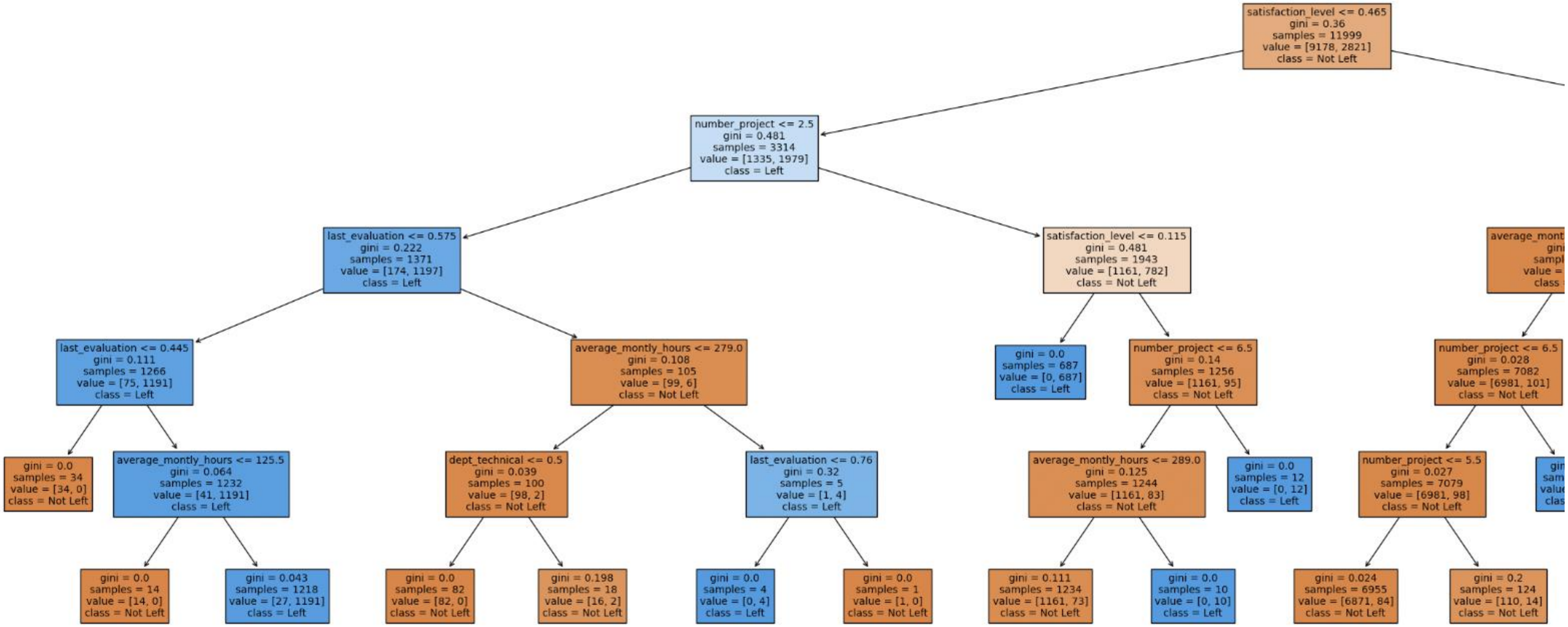


Spam mails that appear in the inbox

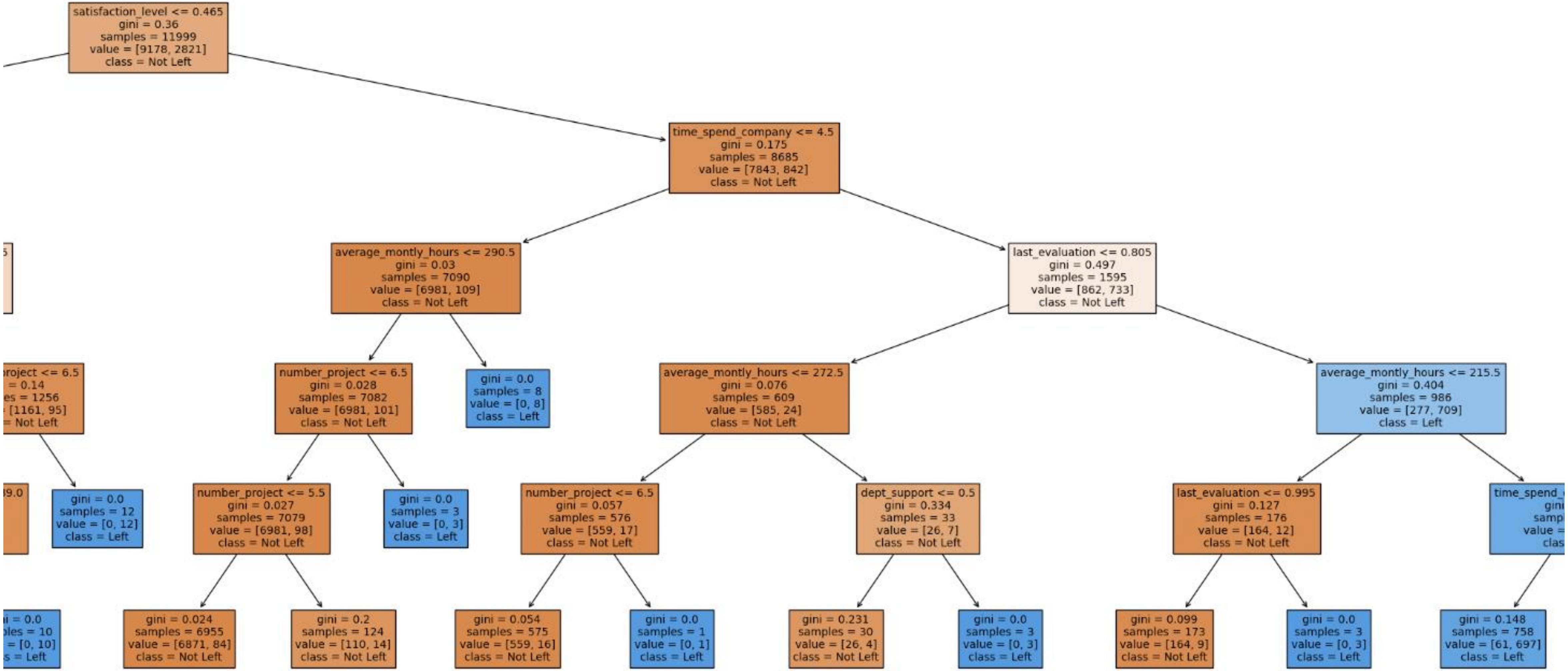
Good mails that appear in the spam box

$$\text{Precision} = \frac{TP}{(TP+FP)}$$

Decision Tree Model



Decision Tree Model



Gini & Entropy

$$Gini = 1 - \sum_{i=1}^{n} p^2(c_i)$$

$$Entropy = \sum_{i=1}^{n} -p(c_i)log_2(p(c_i))$$

where $p(c_i)$ is the probability/percentage of class c_i in a node.

Feature Importance

features	importance	cumsum
satisfaction_level	0.522042	0.522042
time_spend_company	0.158409	0.680451
last_evaluation	0.150507	0.830958
number_project	0.102832	0.933790
average_montly_hours	0.066210	1.000000
Work_accident	0.000000	1.000000
promotion_last_5years	0.000000	1.000000
salary	0.000000	1.000000